## CLASS ROOM VERSION

A salesman travels a distance of 50 km in 2 hours and 30 minutes. How much faster, in kilometers per hour, on an average, must be travel to make such a trip in $\frac{5}{6}$ hour less time?

(a) 10 (b) 20 (c) 30 (d) None of these

An aeroplane flies along the four sides of a square at the speeds of 200, 400, 600 and 800 km/h. Find the average speed of the plane around the field.

(a) 384 km/h (b) 370 km/h
(c) 368 km/h (d) None

A monkey ascends a greased pole 12 metres high. He ascends 2 metres in first minute and slips down 1 metre in the alternate minute. In which minute, he reaches the top?

(a) 21st (b) 22nd (c) 23rd (d) 24th

There are 20 poles with a constant distance between each pole. A car takes 24 seconds to reach the 12th pole. How much time will it take to reach the last pole?

(a) 25.25 s (b) 17. 45 s
(c) 35.75 s (d) 41. 45 s

R and S start walking each other at 10 AM at the speeds of 3 km/h and 4 km/h respectively. They were initially 17.5 km apart. At what time do they meet?

(a) 2:30PM (b) 11:30AM
(c) 1:30PM (d) 12:30PM

A train does a journey without stoppage in 8 hours, if it had travelled 5 km/h faster, it would have done the journey in 6 hours 40 minutes. Find its original speed.

(a) 25km/h (b) 40km/h
(c) 45km/h (d) 36.5 km/h

Cars $C_1$ and $C_2$ travel to a place at a speed of 30 km/h and 45 km/h respectively. If car $C_2$ takes $2\frac{1}{2}$ hours less time than $C_1$ for the journey, the distance of the place is

(a) 300 km (b) 400 km
(c) 350 km (d) 225 km

8. If a man walks to his office at $\frac{3}{4}$ of his usual rate, he reaches office $\frac{1}{3}$ of an hour later than usual. What is his usual time to reach office.

(a) $\frac{1}{2}$ hr (b) 1 hr
(c) $\frac{3}{4}$ hr (d) None of these

9. A car covers 420 km with a constant speed. If its speed were 10 km/h more it would have taken one hour less to cover the distance. Find the speed of the car.

(a) 60km/h (b) 55km/h
(c) 50km/h (d) 48km/h

10. The speed of a car increases by 2 km/h after every one hour. If the distance travelled in the first one hour was 35 kms, what was the total distance travelled in 12 hours?

(a) 456kms (b) 482kms
(c) 552kms (d) 556kms

11. Excluding stoppages, the speed of a train in 45 km/h and including stoppages, it is 36 km/h. For how many minutes does the train stop per hour?

(a) 10 min. (b) 12 min.
(c) 15 min. (d) 18 min.

12. A man travelled a distance of 50 km on his first trip. On a later trip, travelling three times as fast, he covered 300 km. Compare the times he took.

(a) 1 : 2 (b) 2 : 3 (c) 3 : 4 (d) 7 : 9

13. The driver of a car driving @ 36kmph locates a bus 40 meters ahead of him. After 20 seconds the bus is 60 meters behind. The speed of the bus is:

(a) 36kmph (b) 20m/sec
(c) 72m/sec (d) 18 kmph

14. Water drops fall at a uniform rate from a kalash (pot) over the Shiv Ling in the Shiva temple. If 9 drops fall in 10 seconds, how many drops falls in 15 second?
(a) 14 (b) 12 (c) 13 (d) 13.5

15. A train 108 m long moving at a speed of 50 km/h crosses a train 112 m long coming from the opposite direction in 6 seconds. The speed of the second train is
(a) 48km/h (b) 54km/h
(c) 66km/h (d) 82km/h

16. A train 100 metres long takes $3\frac{3}{5}$ seconds to cross a man walking at the rate of 6 km/h in a direction opposite to that of train. Find the speed of the train.
(a) 94 km/hr (b) 100 km/hr
(c) 110 km/hr (d) 108 km/hr

17. The speed of a boat in still water is 15 km/h and the rate of stream is 5 km/h. The distance travelled downstream in 24 minutes is
(a) 4 km (b) 8 km
(c) 6 km (d) 16 km

18. A boat running downstream covers a distance of 16 km in 2 hours while for covering the same distance upstream, it takes 4 hours. What is the speed of the boat in still water?
(a) 4km/h (b) 6km/h
(c) 8km/h (d) Data inadequate

19. A motor boat whose speed is 15 km/h in still water goes 30 km downstream and comes back in four and a half hours. The speed of the stream is:
(a) 46 km/h (b) 6 km/h
(c) 7 km/h (d) 5 km/h

20. A man can row upstream 36 km in 6 hours. If the speed of man in still water is 8 km/h, find how much he can go downstream in 10 hours.
(a) 100 km (b) 110 km
(c) 90 km (d) 115 km

21. A boat takes 90 minutes less to travel 36 miles downstream than to travel the same distance upstream. If the speed of the boat in still water is 10 mph, the speed of the stream is:
(a) 4 mph (b) 3 mph
(c) 2.5 mph (d) 2 mph

22. The rowing speed of man is still water is 20 km/h. Going downstream, he moves at the rate of 25 km/h. The rate of stream is
(a) 45km/h (b) 2.5km/h
(c) 5km/h (d) 10km/h

23. A boat goes 24 km upstream and 28 km downstream in 6 hours. It goes 30 km upstream and 21 km downstream in 6 hours and 30 minutes. The speed of the stream is:
(a) 10 km/h (b) 5 km/h
(c) 4 km/h (d) None

24. A man can row 6 km/hr. in still water. It takes him twice as long to row up as to row down the river. Find the rate of stream
(a) 2 km/hr (b) 4 km/hr
(c) 6 km/hr (d) 8 km/hr

25. A man can swim with the stream at the rate of 3 kmph and against the stream at the rate of 2 kmph. How long will it take him to swim 7 km in still water
(a) 3 hrs (b) 2.8 hrs
(c) 2.6 hrs (d) 3.2 hrs

26. Two trains leave New Delhi at the same time. One travels north at 60 kmph and the other travels south at 40 kmph. After how many hours will the trains be 150 km apart?
(a) $\frac{3}{2}$ (b) $\frac{4}{3}$ (c) $\frac{3}{4}$ (d) $\frac{15}{2}$

27. If a bus travels 160 km in 4 hours and a train travels 320 km in 5 hours at uniform speeds, then what i

the ratio of the distances travelled by them in one hour?

(a) 8 : 5 (b) 5 : 8
(c) 4 : 5 (d) 1 : 2

A man can row at 5 km/hr in still water. If the river is running at 1 km/hr it takes him 75 minutes to row a place and back. How far is the place?

(a) 2.5 km (b) 3 km
(c) 4 km (d) 5 km

A 320 metre long train croses a pole in 16 seconds. It stops five times of duration 18 minutes each. What time will it take in covering a distance of 576 km?

(a) 9 hrs (b) $9\frac{1}{4}$ hrs
(c) $9\frac{1}{2}$ hrs (d) $8\frac{1}{2}$ hrs

A car travels a distance of 560 km in 9.5 hours partly at a speed of 40 $kmh^{-1}$ and partly at 160 $kmh^{-1}$. What distance it travel at the speed of 160 $kmh^{-1}$?

(a) 120 km (b) 240 km
(c) 320 km (d) 360 km

A man rows 750m in 775 seconds against the stream and returns in $7\frac{1}{2}$ minutes. What is rowing speed in still water?

(a) 4.7 km/hr (b) 4 km/hr
(c) 3.5 km/hr (d) 6 km/hr

Two rockets approach each other, one at 42000 mph and the other at 18000 mph. They start 3256 miles apart. How far are they apart (in miles) 1 minute before impact?

(a) 1628 (b) 1000
(c) 826 (d) 1200

A thief steals a car at 2 : 30 p.m. and drives it at 60 kmph. The theft is discovered at 3 p.m. and the owner sets off in another car at 75 kmph. When will he overtake the thief?

(a) 4:30 p.m. (b) 4:45 p.m. (c) 5 p.m. (d) 5:15 p.m.

34. On a journey across Bombay, a tourist bus averages 10 km/h for 20% of the distance, 30 km/h for 60% of it and 20 km/h for the remainder. The average speed for the whole journey was

(a) 10 km/h (b) 30 km/h
(c) 5 km/h (d) 20 km/h

35. A goods train leaves a station at a certain time and at a fixed speed. After 6 hours, an express train leaves the same station and moves in the same direction at a uniform speed of 90 kmph. This train catches up the goods train in 4 hours. Find the speed of the goods train.

(a) 36kmph (b) 40kmph (c) 30kmph (d) 42kmph

36. A train running between two stations A and B arrives at its destination 10 minutes late when its speed is 50 km/h and 50 minutes late when its speed is 30 km/h. What is the distance between the stations A and B?

(a) 40 km (b) 50 km (c) 60 km (d) 70 km

37. A long distance runner runs 9 laps of a 400 metres track everyday. His timings (in minutes) for four consecutive days are 88, 96, 89 and 87 respectively. On an average, how many metres/minute does the runner cover?

(a) 40m/min (b) 45m/min
(c) 38m/min (d) 49m/min

38. Mohan travels 760 km to his home, partly by train and partly by car. He takes 8 hours if he travels 160 km by train and the rest by car. He takes 12 minutes more if he travels 240 km by train and the rest by car. The speed of the train and the car, respectively are:

(a) 80 km/h, 100 km/h (b) 100 km/h, 80 km/h
(c) 120 km/h, 120 km/h (d) 100 km/h, 120 km/h

39. A walks around a circular field at the rate of one round per hour while B runs around it at the rate of six rounds per hour. They start in the same direction from the same point at 7:30 a.m. They shall first cross each other at:

(a) 7:42 a.m. (b) 7:48 a.m. (c) 8:10 a.m. (d) 8:30 a.m

**40. A circular running path is 726 metres in circumference. Two men start from the same point and walk in opposite directions @ 3.75 km/h and 4.5 km/h, respectively. When will they meet for the first time?**

(a) After 5.5 min (b) After 6.0 min
(c) After 5.28 min (d) After 4.9 min

**41. Local trains leave from a station at an interval of 14 minutes at a speed of 36 km/h. A man moving in the same direction along the road meets the trains at an interval of 18 minutes. Find the speed of the man.**

(a) 8 km/hr (b) 7 km/hr (c) 6 km/hr (d) 5.8 km/hr

**42. Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:**

(a) 1 : 3 (b) 3 : 2 (c) 3 : 4 (d) None of these

**43. Two trains each of 120 m in length, run in opposite directions with a velocity of 40 m/s and 20 m/s respectively. How long will it take for the tail ends of two trains to meet each other during the course of their journey:**

(a) 20 s (b) 3 s (c) 4 s (d) 5 s

**44. A boat running upstream takes 8 hours 48 minutes to cover a certain distance, while it takes 4 hours to cover the same distance running downstream. What is the ratio between the speed of the boat and speed of water current respectively?**

(a) 2 : 1 (b) 3 : 2
(c) 8 : 3 (d) Cannot be determined

**45. A boy rows a boat against a stream flowing at 2 kmph for a distance of 9 km, and then turns round and rows back with the current. If the whole trip occupies 6 hours, find the boy's rowing speed in still water.**

(a) 4 kmph (b) 3 kmph (c) 2 kmph (d) 5 kmph

***End Of Classroom Version Time, Speed & Distance

**STUDENT VERSION**

**6. A can complete a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. Find the total journey in km.**

(a) 220 km (b) 224 km

(c) 230 km (d) 234 km

**. A person can walk a certain distance and drive back in six hours. He can also walk both ways in 10 hours. How much time will he take to drive both ways?**

(a) Two hours (b) Two and a half hours

(c) Five and a half hours (d) Four hours

**. A man is walking at a speed of 10 km per hour. After every kilometre, he takes rest for 5 minutes. How much time will he take to cover a distance of 5 kilometres?**

(a) 48 min (b) 50 min

(c) 45 min (d) 55 min

**If I walk at 4 km/h, I miss the bus by 10 minutes. If I walk at 5 km/h, I reach 5 minutes before the arrival of the bus. How far I walk to reach the bus stand?**

(a) 5 km (b) 4.5 km

(c) $5\frac{1}{4}$ -km/h (d) Cannot be determined

**If a man walks to his office at 5/4 of his usual rate, he reaches office 30 minutes early than usual. What is his usual time to reach office.**

(a) 2 hr (b) $2\frac{1}{2}$ hr

(c) 1 hr 50 min (d) 2 hr 15 min

**. A train travels at an average of 50 miles per hour for $2\frac{1}{2}$ hours and then travels at a speed of 70 miles per hour for $1\frac{1}{2}$ hours. How far did the train travel in the entire 4 hours?**

(a) 120 miles (b) 150 miles

(c) 200 miles (d) 230 miles

**52. A man in a train notices that he can count 21 telephone posts in one minute. If they are known to be 50 metres apart, then at what speed is the train travelling?**

(a) 55 km/hr (b) 57 km/hr

(c) 60 km/hr (d) 63 km/hr

**53. Mary jogs 9 km at a speed of 6 km per hour. At what speed would she need to jog during the next 1.5 hours to have an average of 9 km per hour for the entire jogging session?**

(a) 9 kmph (b) 10 kmph

(c) 12 kmph (d) 14 kmph

**54. Walking $\frac{6}{5}$ of his usual speed, a person takes 10 min less to reach his office. His usual time taken to reach the office is**

(a) 50 min (b) 55 min

(c) 60 min (d) None of these

**55. Mr. A is running at a speed X kmph to cover a distance 1 km. But due to sticky ground his speed is reduced by Y kmph. If he takes Z hours to cover the distance, then -**

(a) $\frac{1}{Z}=\frac{1}{X}+\frac{1}{Y}$ (b) $ZX-ZY=1$

(c) $Z=X+Y$ (d) None of the above

**56. 150 metres long train crosses a bridge of length 250 metres in 30 seconds. Find the time for train to cross a platform of 130 metres.**

(a) 21 seconds (b) 23 seconds

(c) 20 seconds (d) 24 seconds

**57. A train passes a station platform in 36 seconds and a man standing on the platform in 20 sec onds. If the speed of the train is 54 km/h, find the**

length of the platform

(a) 240 metres (b) 220 metres

(c) 258 metres (d) None of these

58. A train 150 m long is running with a speed of 68 kmph. In R what time will it pass a man who is running at 8 kmph in the same direction in which the train is going ?

(a) 10 sec (b) 9 sec

(c) 8 sec (d) 6 sec

59. A person can swim in still water at 4 km/h. If the speed of water is 2 km/h, how many hours will the man take to swim back against the current for 6km.

(a) 3 (b) 4

(c) $4\frac{1}{2}$ (d) Insufficient data

60. A man can row 4 km/h in still water and he finds that it takes him twice as long to row up as to row down the river. Find the rate of stream.

(a) 1.3 km/h (b) 1.5 km/h

(c) 1.8 km/h (d) None of these

61. A man can row 60 km down stream in 6 hours. If the speed of the current is 3 km/h, then find in what time will he be able to cover 16 km up stream?

(a) 5 hrs (b) 4 hrs

(c) 3 hrs (d) None of these

62. A man can swim in still water at 4.5 km/h, but takes twice as long to swim upstream than down stream. The speed of the stream is

(a) 3 (b) 7.5 (c) 2.25 (d) 1.5

63. A man makes his upward journey at 16 km/h and downward journey at 28 km/h. What is his aver age speed?

(a) 32 km/h (b) 56 km/h

(c) 20 36 km/h (d) 22 km/h

64. A man rows to a place 48 km distant and back i...



14 hours. He finds that he can row 4 km with the stream in the same time as 3 km against the stream. Find the rate of stream."

(a) 1 km/h (b) 1.5 km/h

(c) 2 km/h (d) None of these

65. A boat covers a certain distance downstream in 1 hour, while it comes back in $1\frac{1}{2}$ hours. If the speed of the stream be 3 kmph, what is the speed of the boat in still water?

(a) 12 kmph (b) 13 kmph

(c) 14 kmph (d) 15 kmph

66. Against a stream running at 2 km an hour, a man can row 9 km in 3 hours. How long would he take in rowing the same distance down the stream?

(a) 9/7 hours (b) 7/9 hours

(c) 1.5 hours (d) 3 hours

67. A man can swim with the stream at the rate of 10.56 kmph and 352 m against the stream in 4 minutes. The speed of the man in still water is:

(a) 10.56 kmph (b) 352 m/minutes

(c) 7.92 kmph (d) 5.28 kmph

68. Running at a speed of 60 km per hour, a trai passed through a 1.5 km long tunnel in two minutes. What is the length of the train?

(a) 250m (b) 500m

(c) 1000m (d) 1500m

69. A boat covers 18 km downstream in 3 hours. If speed of current is $33\frac{1}{3}$% of its downstream speed. In what time will it cover a distance of 50 km upstream?

(a) 50 hour (b) 40 hour

(c) 25 hour (d) 60 hour

70. An inspector is 228 meter b... inspecto...

meters in a minute. In what time will the inspector catch the thief?

(a) 19 minutes (b) 20 minutes
(c) 18 minutes (d) 21 minutes

71. A is twice as fast as B and B is trice as fast as C is the journey covered by C in $1\frac{1}{2}$ hours will be covered by A in

(a) 15 minutes (b) 2 minutes
(c) 30 minutes (d) 1 hour

72. In a 800 m race around a stadium having the circumference of 200 m, the top runner meets the last runner on the 5th minute of the race. If the top runner runs at twice the speed of the last runner, what is the time taken by the top runner to finish the race?

(a) 20 min (b) 15 min
(c) 10 min (d) 5 min

73. A man covers a certain distance on a toy train. If the train moved 4 km/h faster, it would take 30 minutes less. If it moved 2 km/h slower, it would have taken 20 minutes more. Find the distance.

(a) 60 km (c) 55 km
(b) 58 km (d) 50 km

74. A train completes a journey with a few stoppages in between at an average speed of 40 km per hour. If the train had not stopped anywhere, it would have completed the journey at an average speed of 60 km per hour. On an average, how many minutes per hour does the train stop during the journey?

(a) 20 minutes per hour (b) 18 minutes per hour
(c) 15 minutes per hour (d) 10 minutes per hour

75. A car travels 25 km an hour faster than a bus for a journey of 500 km. If the bus takes 10 hours more than the car, then the speeds of the bus and the car are

(a) 25 km/h and 40 km/h respectively
(b) 25 km/h and 60 km/h respectively
(c) 25 km/h and 50 km/h respectively
(d) None of these

75. A person has to cover a distance of 6 km in 45 minutes. If he covers one-half of the distance in two-thirds of the total time; to cover the remaining distance in the remaining time, his speed (in km/h) must be:

(a) 6 (b) 8 (c) 12 (d) 15

76. A thief steals a car at 2 : 30 p.m. and drives it at 60 kmph. The theft is discovered at 3 p.m. and the owner sets off in another car at 75 kmph. When will he overtake the thief?

(a) 4:30 p.m. (b) 4:45 p.m. (c) 5 p.m. (d) 5:15 p.m.

77. A train travels at a certain average speed for a distance of 225 km and then travels a distance of 245 km at an average speed of 5 km/hr more than its original speed. If it takes 5 hours to complete the total journey, what is the original speed of the train in km/hr.

(a) 88 (b) 90 (c) 80 (d) 85

78. A plane left 30 minutes later than the scheduled time and in order to reach the destination 1500 km away in time, it had to increase the speed by 250 km/h from the usual speed. Find its usual speed.

(a) 720 km/h (b) 740 km/h
(c) 730 km/h (d) 750 km/h

79. A train X starts from Meerut at 4 p.m. and reaches Ghaziabad at 5 p.m. while another train Y starts from Ghaziabad at 4 p.m. and reaches Meerut at 5.30 p.m. The two trains will cross each other at:

(a) 4:36 p.m. (b) 4:42 p.m.
(c) 4:48 p.m. (d) 4:50 p.m.

80. A train consists of 12 boggies, each boggy 15 metres long. The train crosses a telegraph post in 18 seconds. Due to some problem, two boggies were detached. The train now crosses a telegraph post in

(a) 18 sec (b) 12 sec (c) 15 sec (d) 20 sec

81. A man sitting in a train which is travelling at 50 kmph observes that a goods train, travelling in opposite direction, takes 9 seconds to pass him. If the goods train is 280 m long, find its speed.
(a) 62 kmph (b) 58 kmph (c) 52 kmph (d) None

82. A train 75 metres long overtook a man who was walking at the rate of 6 km/h and passed him in 18 seconds. Again, the train overtook a second person in 15 seconds. At what rate was the second person travelling?
(a) 3 km/h (b) 2.5 km/h (c) 4 km/h (d) None

83. Two trains are running at 40 km/h and 20 km/h respectively in the same direction. Fast train completely passes a man sitting in the slower train in 5 seconds. What is the length of the fast train?
(a) 23 m (b) $23\frac{2}{9}$ m (c) 27 m (d) $27\frac{7}{9}$ m

84. Two trains, 130 and 110 metres long, while going in the same direction, the faster train takes one minute to pass the other completely. If they are moving in opposite direction, they pass each other completely in 3 seconds. Find the speed of trains.
(a) 30 m/s, 40 m/s (b) 32 m/s, 48 m/s
(c) 40 m/s, 44 m/s (d) 38 m/s, 42 m/s

85. Mr. Kumar drives to work at an average speed of 48 km per hour. The time taken to cover the first 60% of the distance is 10 minutes more than the time taken to cover the remaining distance. How far is his office?
(a) 30 km (b) 40 km (c) 45 km (d) 48 km

86. A train travels at a certain average speed for a distance of 63 km and then travels a distance of 72 km at an average speed of 6 km/h more than its original speed. If it takes 3 hours to complete the total journey, what is the original speed of the train in km/hr?
(a) 24 (b) 33 (c) 42 (d) 66

87. Location of B is north of A and location of C is ... of A. The distances AB and AC are 5 km and 12 km respectively. The shortest distance (in km) between the locations B and C is
(a) 60 (b) 13 (c) 17

88. A worker reaches his factor 3 minutes late if his speed from his house to the factory is 5 km/hr. ... he walks at a speed of 6 km/hr, then he reaches the factory 7 minute early. The distance of the factory from his house is
(a) 3 km (b) 4 km (c) 5 km (d) 6 ...

89. Sum of the length of two trains P and Q is 330. The ratio of the speeds of P and Q is 5 : 8. Ratio between time to cross an electric pole by P and Q is 4 : 3. Find the difference in the length of two trains.
(a) 50 (b) 30 (c) 40 (d) 75

90. A train is 216 m long. It crosses a platform in 19 seconds with speed 21 m/s. If some 21 m long coaches are added in train and it crosses same platform, then it takes 26 seconds to cross the platform at same speed. How many coaches were added to the train?
(a) 7 (b) 10 (c) 12 (d)

91. Ashok left from place A (towards place B) at 8 am and Rahul left from place B (towards place A) at 1... am. The distance between place A and place B is 637 km. If Ashok and Rahul are travelling at a uniform speed of 39 km/h and 47 km/h respectively at what time will they meet?
(a) 5:30 pm (b) 4:30 pm (c) 5 pm (d) 4 pm

92. In a 500 meters race, B starts 45 meters ahead of A but A wins the race while B is still 35 meters behind. What is the ratio of the speeds of A to B assuming that both start at the same time?
(a) 12.5 km (b) 10 km (c) 15 km (d) 18 km

93. Two cities A and B are 360 km apart. A car goes from A to B with a speed of 40 km/hr and returns to A with a speed of 60 km/hr. What is the average speed of the car?

(a) 45 km/hr (b) 48 km/hr (c) 50 km/hr (d) 55 km/hr

4. A and B walk around a circular park. They start at 8 a.m. from the same point in the opposite directions. A and B walk at a speed of 2 rounds per hour and 3 rounds per hour respectively. How many times shall they cross each other after 8:00 a.m. and before 9:30 a.m.?

(a) 7 (b) 6 (c) 5 (d) 8

. Four friends A, B, C and D need to cross a bridge. A maximum of two persons can cross it at a time. It is night and they just have one lamp. Persons that cross the bridge must carry the lamp to find the way. A pair must walk together at the speed of slower person. After crossing the bridge, the person having faster speed in the pair will return with the lamp each time to accompany another person in the group. Finally, the lamp has to be returned at the original place and the person who returns the lamp has to cross the bridge again without lamp. To cross the bridge, the time taken by them is as follows: A : 1 minute, B : 2 minutes, C : 7 minutes and D : 10 minutes. What is the total minimum time required by all the friends to cross the bridge?

(a) 23 minutes (b) 22 minutes
(c) 21 minutes (d) 20 minutes

96. A daily train is to be introduced between station A and station B starting from each at 6 AM and the journey is to be completed in 42 hours. What is the number of trains needed in order to maintain the Shuttle Service?

(a) 2 (b) 3 (c) 4 (d) 7

97. A freight train left Delhi for Mumbai at an average speed of 40 km/hr. Two hours later, an express train left Delhi for Mumbai, following the freight train on a parallel track at an average speed of 60 km/hr. How far from Delhi would the express train meet the freight train?

(a) 480 km (b) 260 km (c) 240 km (d) 120 km

98. A train 200 meters long is moving at the rate of 40 kmph. In how many seconds will it cross a man standing near the railway line?

(a) 12 (b) 15 (c) 16 (d) 18

99. Two persons, A and B are running on a circular track. At the start, B is ahead of A and their positions make an angle of 30° at the centre of the circle. When A reaches the point diametrically opposite to his starting point, he meets B. What is the ratio of speeds of A and B, if they are running with uniform speeds?

(a) 6 : 5 (b) 4 : 3 (c) 6 : 1 (d) 4 : 2

***End Of Student Version Time,Speed & Distance***

## PYQs

0. Two stations A and B were 310 km apart on a straight line. One train starts from A at 10 a.m. and travels towards B at 40 km/hr. Another train starts from B at 11 a.m. and travels towards A at a speed of50 km/hr. At what time will they meet?

(a) 3 P.M. (b) 4 P.M. (c) 12 P.M.
(d) 2 P.M. (e) 2 : 30 P.M.

01. The speed of train A and train B is 93 km/hr and 51 km/hr, respectively. When both the trains are running in the opposite direction, they cross each other in 18 seconds. The length of train B is half of the length of train A. If train A crosses a bridge in 42 seconds, then find the length of the bridge.

(a) 610 m (b) 480 m (c) 605 m
(d) 240 m (e) 485 m

102. The speed of a boat in still water is 6 km/h. Time taken by the boat to cover a certain distance upstream is 3 hours more than the time taken to cover the same distance downstream. If the

speed of the stream is 2 km/h, then what is the total distance (upstream and downstream) covered by the boat?

(a) 24 km (b) 18 km (c) 36 km
(d) 48 km (e) 46 km

**103.** Train A crosses a stationary train B in 90 sec -onds and a pole in 30 seconds with the same speed.The length of the train A is 270 metres. What is the length of the stationary train B?

(a) 540 meters (b) 270 meters
(c) 810 meters (d) cannot be determined
(e) None of these

**104. A motorboat** stream in 2 h If the speed speed of boa

(a) 2.71 m/s
(c) 13 km/hr

**105. A train runni** platform in 30 and its length running at sa 46seconds. Fi

(a) 180 m (b)
(d) 240 m (e)